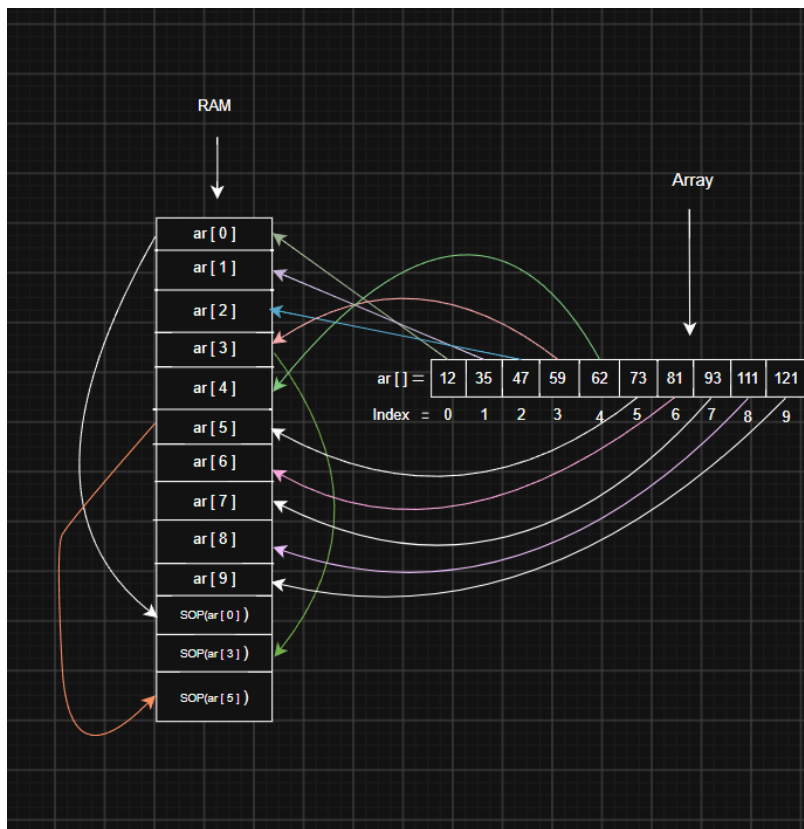


Q01.

i.

```
class Q1 {  
    public static void main(String[] args){  
        int[] ar={12,35,47,59,62,73,81,93,111,121};  
    }  
}
```

ii,iii.



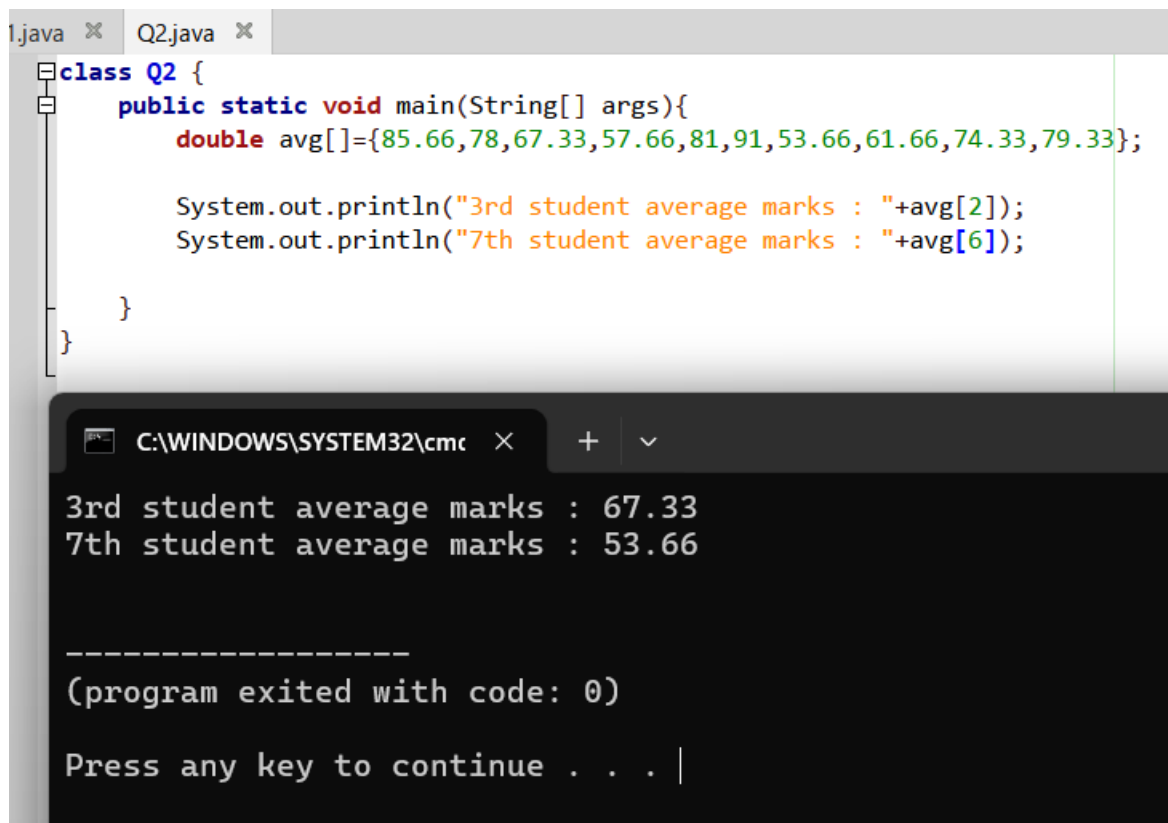
iv.

```
File Edit Search View Document Project Build Tools Help  
Q1.java x  
1 class Q1 {  
2     public static void main(String[] args){  
3         int[] ar={12,35,47,59,62,73,81,93,111,121};  
4  
5         System.out.println(ar[0]);  
6         System.out.println(ar[3]);  
7         System.out.println(ar[5]);  
8     }  
9 }  
10
```

C:\WINDOWS\SYSTEM32\cmd x + v

```
12  
59  
73  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

Q02.



```
1.java x Q2.java x
class Q2 {
    public static void main(String[] args){
        double avg[]={85.66,78,67.33,57.66,81,91,53.66,61.66,74.33,79.33};

        System.out.println("3rd student average marks : "+avg[2]);
        System.out.println("7th student average marks : "+avg[6]);
    }
}

C:\WINDOWS\SYSTEM32\cmd x + v
3rd student average marks : 67.33
7th student average marks : 53.66

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q03.

i,ii,iii,iv

```
class Q3 {
    public static void main(String[] args){
        String month[]={"January","February","March","April","May","June","July","August","September","October","November","December"};
        int exchange[]={2362,2311,1917,1912,1887,1854,1717,1717,1779,1705,1806,1896};

        System.out.println("        Sri lanka        ");
        System.out.println(" Foreign Exchange Reserves ");
        System.out.println("-----");
        System.out.println("|Month        |USD Milion |");
        System.out.println("-----");
        System.out.println("|"+month[0]+"      |          "+exchange[0]+"|");
        System.out.println("-----");
        System.out.println("|"+month[1]+"      |          "+exchange[1]+"|");
        System.out.println("-----");
        System.out.println("|"+month[2]+"      |          "+exchange[2]+"|");
        System.out.println("-----");
        System.out.println("|"+month[3]+"      |          "+exchange[3]+"|");
        System.out.println("-----");
        System.out.println("|"+month[4]+"      |          "+exchange[4]+"|");
        System.out.println("-----");
        System.out.println("|"+month[5]+"      |          "+exchange[5]+"|");
        System.out.println("-----");
        System.out.println("|"+month[6]+"      |          "+exchange[6]+"|");
        System.out.println("-----");
        System.out.println("|"+month[7]+"      |          "+exchange[7]+"|");
        System.out.println("-----");
        System.out.println("|"+month[8]+"      |          "+exchange[8]+"|");
        System.out.println("-----");
        System.out.println("|"+month[9]+"      |          "+exchange[9]+"|");
        System.out.println("-----");
        System.out.println("|"+month[10]+"     |          "+exchange[10]+"|");
        System.out.println("-----");
        System.out.println("|"+month[11]+"     |          "+exchange[11]+"|");
        System.out.println("-----\n");

        System.out.println(" Modify (April and july)\n");

        exchange[3]=1812;
        exchange[6]=1817;

        System.out.println("-----");
        System.out.println("|Month        |USD Milion |");
        System.out.println("-----");
        System.out.println("|"+month[3]+"      |          "+exchange[3]+"|");
        System.out.println("-----");
        System.out.println("|"+month[6]+"      |          "+exchange[6]+"|");
        System.out.println("-----");
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v

Sri lanka
Foreign Exchange Reserves
-----
|Month      |USD Milion |
-----
|January    |      2362|
-----
|February   |      2311|
-----
|March      |      1917|
-----
|April      |      1912|
-----
|May        |      1887|
-----
|June       |      1854|
-----
|July       |      1717|
-----
|August     |      1717|
-----
|September  |      1779|
-----
|October    |      1705|
-----
|November   |      1806|
-----
|December   |      1896|
-----

Modify (April and july)

-----
|Month      |USD Milion |
-----
|April      |      1812|
-----
|July       |      1817|
-----

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q04.

I,ii

```
1.java ✕ Q2.java ✕ Q3.java ✕ Q4.java ✕
class Q4 {
    public static void main(String[] args){

        char[] ar={'I','n','s','t','i','t','u','t','e',' ','O','f',' ','C','o','m','p','u','t','e','r'
                    ,',' ','E','n','g','i','n','e','e','r','i','n','g',' ','T','e','c','h','n','o','l','o','g','y'};

        System.out.println("'" + ar[0] + ar[1] + ar[2] + ar[3] + ar[4] +
            ar[5] + ar[6] + ar[7] + ar[8]);

        System.out.println("'" + ar[10] + ar[11]);

        System.out.println("'" + ar[13] + ar[14] + ar[15] + ar[16] +
            ar[17] + ar[18] + ar[19] + ar[20]);

        System.out.println("'" + ar[22] + ar[23] + ar[24] + ar[25] + ar[26] + ar[27] +
            ar[28] + ar[29] + ar[30] + ar[31] + ar[32]);

        System.out.println("'" + ar[34] + ar[35] + ar[36] + ar[37] +
            ar[38] + ar[39] + ar[40] + ar[41] + ar[42] + ar[43]);
    }
}
```

Output>>

```
I.java
Institute
Of
Computer
Engineering
Technology

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q05.

i,ii

```
1.java  Q2.java  Q3.java  Q4.java  Q5.java
import java.util.*;
class Q5 {
    public static void main(String[] args){
        int marks[]={56,81,43,69,93,54,48,47,51,79,82,96,57,61,66};

        System.out.println("There are "+marks.length+" marks input into the array");

        //for(int i=0; i<5; i++){
        //    System.out.println(marks[i]);
        //}

        //for(int i=0; i<marks.length; i++){
        //    System.out.println(marks[i]);
        //}
        System.out.println(marks[9]);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
There are 15 marks input into the array

-----
(program exited with code: 0)
```

iii.

```
1.java  Q2.java  Q3.java  Q4.java  Q5.java
import java.util.*;
class Q5 {
    public static void main(String[] args){
        int marks[]={56,81,43,69,93,54,48,47,51,79,82,96,57,61,66};

        //System.out.println("There are "+marks.length+" marks input into the array");

        for(int i=0; i<5; i++){
            System.out.println(marks[i]);
        }

        //for(int i=0; i<marks.length; i++){
        //    System.out.println(marks[i]);
        //}
        System.out.println(marks[9]);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
Q1.j:
1 56
2 81
3 43
4 69
5 93
6
7
8 -----
9 (program exited with code: 0)
10
```

iv.

```
import java.util.*;
class Q5 {
    public static void main(String[] args){
        int marks[]={56,81,43,69,93,54,48,47,51,79,82,96,57,61,66};

        //System.out.println("There are "+marks.length+" marks input into the array");

        //for(int i=0; i<5; i++){
            //System.out.println(marks[i]);
        //}

        for(int i=0; i<marks.length; i++){
            System.out.println(marks[i]);
        }
        //System.out.println(marks[9]);
    }
}
```

```
56
81
43
69
93
54
48
47
51
79
82
96
57
61
66

-----
(program exited with code: 0)
Press any key to continue . . . |
```

v.

```
Q1.java Q2.java Q3.java Q4.java Q5.java
1 import java.util.*;
2 class Q5 {
3     public static void main(String[] args){
4         int marks[]={56,81,43,69,93,54,48,47,51,79,82,96,57,61,66};
5
6         //System.out.println("There are "+marks.length+" marks input into the array");
7
8         //for(int i=0; i<5; i++){
9             //System.out.println(marks[i]);
10        //}
11
12        //for(int i=0; i<marks.length; i++){
13            //System.out.println(marks[i]);
14        //}
15        System.out.println(marks[9]);
16    }
17 }
18 }
```

```
79

-----
```

Q06.

```
import java.util.*;
class Q6 {
    public static void main(String[] args){
        String days[]={"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};

        System.out.print(Arrays.toString(days));

        System.out.println();
        System.out.print("[");
        for(int i = days.length - 1; i >=0; i--) {
            System.out.print(days[i]+" ");
            if(i>0){
                System.out.print(",");
            }
        }
        System.out.print("]");
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
[Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday]
[Saturday ,Friday ,Thursday ,Wednesday ,Tuesday ,Monday ,Sunday ]
-----
(program exited with code: 0)
```

Q07.

```
import java.util.*;

class Q7 {
    public static void main(String[] args){
        char[] ar ={'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

        System.out.println(Arrays.toString(ar)+"\n");

        for(int i=0;i<ar.length;i+=2){
            System.out.print(ar[i]+" ");
        }
        System.out.println("\n");
        for(int i=1;i<ar.length;i+=2){
            System.out.print(ar[i]+" ");
        }
        System.out.println("\n");
        char[] rever=new char[ar.length];
        for(int i=0 ,z=ar.length-1;i<ar.length;i++,z--){

            rever[i]=ar[z];
        }
        System.out.print(Arrays.toString(rever));
    }
}
```

Output>>

```
[A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z]
A C E G I K M O Q S U W Y
B D F H J L N P R T V X Z
[Z, Y, X, W, V, U, T, S, R, Q, P, O, N, M, L, K, J, I, H, G, F, E, D, C, B, A]
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q08.

```
import java.util.*;
class Q8 {
    public static void main(String[] args){
        int districts[]={2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903, 12061, 1005, 1545,
        2156, 2037, 1583, 3668};

        System.out.println("Number of Districts : "+districts.length);
        int morethan_count =0;
        int lessthan_count =0;
        for(int i=0;i<districts.length;i++){
            if(districts[i]>2000){
                morethan_count++;
            }
        }
        System.out.println("Nmbers of districts with more than 2000mm of annual rainfall : "+morethan_count);

        for(int i=0;i<districts.length;i++){
            if(districts[i]<1000){
                lessthan_count++;
            }
        }
        System.out.println("Nmbers of districts with less than 1000mm of annual rainfall : "+lessthan_count);
        int sum=0;
        for(int i=0;i<districts.length;i++){
            sum=sum+districts[i];
        }
        double rain_avg =(double)sum/districts.length;
        System.out.printf("The average annual rainfall values of all districs : %.2f",rain_avg);
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd  X + v
Number of Districts : 18
Nmbers of districts with more than 2000mm of annual rainfall : 8
Nmbers of districts with less than 1000mm of annual rainfall : 0
The average annual rainfall values of all districs : 2410.06

-----
(program exited with code: 0)
```

Q09.

09. Which can be inserted at line 12, still the code will compile?

```
class Example{  
    public static void main(String args[]){  
        int[] array;  
        //Insert code here //Line 12  
    }  
}
```

A. array=new int[5];

B. array=new int[10];

C. array=new int[-5];

D. array={10, 20, 30, 40, 50};

E. array=new int[] {10, 20, 30, 40, 50};

F. array=new int[] {};

Answers:

A,B,C,E,F

Q10.

```
Q1.java × Q2.java × Q3.java × Q4.java × Q5.java × Q6.java × Q7.java × Q8.java × Q9.java × Q10.java ×
1 class Q10 {
2     public static void main(String[] args){
3         int result[]={6,3,3,2,4,1,6,6,2,4};
4         int count=0;
5
6         for(int i=0;i<result.length;i++){
7             if(result[i]==3 | result[i]==6){
8                 count++;
9             }
10        }
11        System.out.println("the number of times '3' and '6' is rolled : "+count);
12        int count_odd=0;
13        for(int i=0;i<result.length;i++){
14            if(result[i]%2==1){
15                count_odd++;
16            }
17        }
18        System.out.println("the number of times odd numbers are rolled : "+count_odd);
19        boolean found_5=false;
20        for(int i=0;i<result.length;i++){
21            if(result[i]==5){
22                found_5=true;
23                break;
24            }
25        }
26        System.out.println("Is number '5' are rolled : "+(found_5 ? "Yes":"No"));
27    }
28 }
```

Output>>

```
the number of times ?3? and ?6? is rolled : 5
the number of times odd numbers are rolled : 3
Is number ?5? are rolled : No
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

Q11.

i.

```
class Q11 {  
    public static void main(String[] args){  
        int[] xr=new int[3];  
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4  
    }  
}
```

```
0 0 0  
-----  
(program exited with code: 0)
```

Explain:without array initialization (without values)

It prints default output (0).

ii.

char

```
class Q11 {  
    public static void main(String[] args){  
        char[] xr=new char[3];  
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4  
    }  
}
```

```
C:\WINDOWS\SYSTEM32\cmd  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

boolean

```
Q1.java Q2.java Q3.java Q4.java Q5.java Q6.java Q7.java Q8.java
1 class Q11 {
2     public static void main(String[] args){
3         boolean[] xr=new boolean[3];
4         System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
5     }
6 }
7 }
8 }
```

```
C:\WINDOWS\SYSTEM32\cmd
false false false

-----
(program exited with code: 0)
```

double

```
Q1.java Q2.java Q3.java Q4.java Q5.java Q6.java Q7.java Q8.java
1 class Q11 {
2     public static void main(String[] args){
3         double[] xr=new double[3];
4         System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
5     }
6 }
7 }
8 }
```

```
C:\WINDOWS\SYSTEM32\cmd
0.0 0.0 0.0

-----
(program exited with code: 0)
```

String

```
class Q11 {
    public static void main(String[] args){
        String[] xr=new String[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
null null null

-----
(program exited with code: 0)
```

Q12.

```
Q1.java × Q2.java × Q3.java × Q4.java × Q5.java × Q6.java × Q7.java × Q8.java
import java.util.*;
class Q12 {
    public static double markTotal=0;
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        double ar[]= new double[9];

        for(int i=0;i<ar.length;i++){

            System.out.print("Enter Student Subject "+(i+1)+" marks : ");
            ar[i]=input.nextDouble();
        }
        for(int i =0;i<ar.length;i++){
            markTotal+=ar[i];
        }
        System.out.println("The total mark of the student : "+markTotal);
        System.out.printf("The average mark of the student : %.2f", (markTotal/ar.length));
        System.out.println();
        double max = 0;
        double min = ar[0];
        for(int i=1;i<ar.length;i++){
            if(ar[i]<=min){
                min=ar[i];
            }
        }
        System.out.println("The minimum mark of the student : "+min);
        for(int i=0;i<ar.length;i++){
            if(ar[i]>=max){
                max=ar[i];
            }
        }
        System.out.println("The maximum mark of the student : "+max);
        int count_g=0;
        for(int i =0;i<ar.length;i++){

            if(ar[i]>75){
                count_g++;
            }
        }
        System.out.println("How many marks are greater than 75 : "+count_g);
        int count_l=0;
        for(int i =0;i<ar.length;i++){

            if(ar[i]<45){
                count_l++;
            }
        }
        System.out.println("How many marks are less than 45 : "+count_l);
        boolean found_100=false;
        for(int i=0;i<ar.length;i++){
            if(ar[i]==100){
                found_100=true;
                break;
            }
        }
        System.out.println("Is number '5' are rolled : "+(found_100? "Yes":"No"));
    }
}
```

```
Enter Student Subject 1 marks : 67
Enter Student Subject 2 marks : 5
Enter Student Subject 3 marks : 68
Enter Student Subject 4 marks : 7
Enter Student Subject 5 marks : 98
Enter Student Subject 6 marks : 3
Enter Student Subject 7 marks : 100
Enter Student Subject 8 marks : 5
Enter Student Subject 9 marks : 46
The total mark of the student      : 399.0
The average mark of the student    : 44.33
The minimum mark of the student    : 3.0
The maximum mark of the student    : 100.0
How many marks are greater than 75 : 2
How many marks are less than 45    : 4
Is number 75? are rolled : Yes
```

(program exited with code: 0)

Q13.

i.

A.has Scanner method to get input by user

B.has Random method to get random numbers by System.

ii.

```
numer[i]=r.nextInt(101);
```

Q14.

```
Q1.java ✕ Q2.java ✕ Q3.java ✕ Q4.java ✕ Q5.java ✕ Q6.java ✕ Q7.java ✕ Q8.java ✕ Q9.java ✕ Q10.java ✕ Q11.java ✕
1  import java.util.*;
2  class Q14 {
3      public static void main(String[] args){
4          Random r = new Random();
5          int[] ar= new int[20];
6
7          for(int i=0;i<ar.length;i++){
8              ar[i]= r.nextInt(101);
9
10         }
11         System.out.println(Arrays.toString(ar));
12         int pass_count=0;
13         for(int i=0;i<ar.length;i++){
14             if(ar[i]>50){
15                 pass_count++;
16             }
17         }
18         System.out.println("\n");
19         System.out.println("How many students pass the assignmen : "+pass_count);
20
21         boolean found90=false;
22         for(int i=0;i<ar.length;i++){
23             if(ar[i]>90){
24                 found90=true;
25                 break;
26             }
27         }
28         System.out.println("Are there student with more than 90 : "+(found90 ? "Yes":"No"));
29     }
30 }
31
```

Output>>

```
[39, 8, 48, 12, 91, 71, 61, 13, 5, 48, 4, 84, 33, 95, 73, 82, 79, 26, 2, 87]
```

```
How many students pass the assignmen : 9
Are there student with more than 90 : Yes
```

```
-----
(program exited with code: 0)
```

Q15.

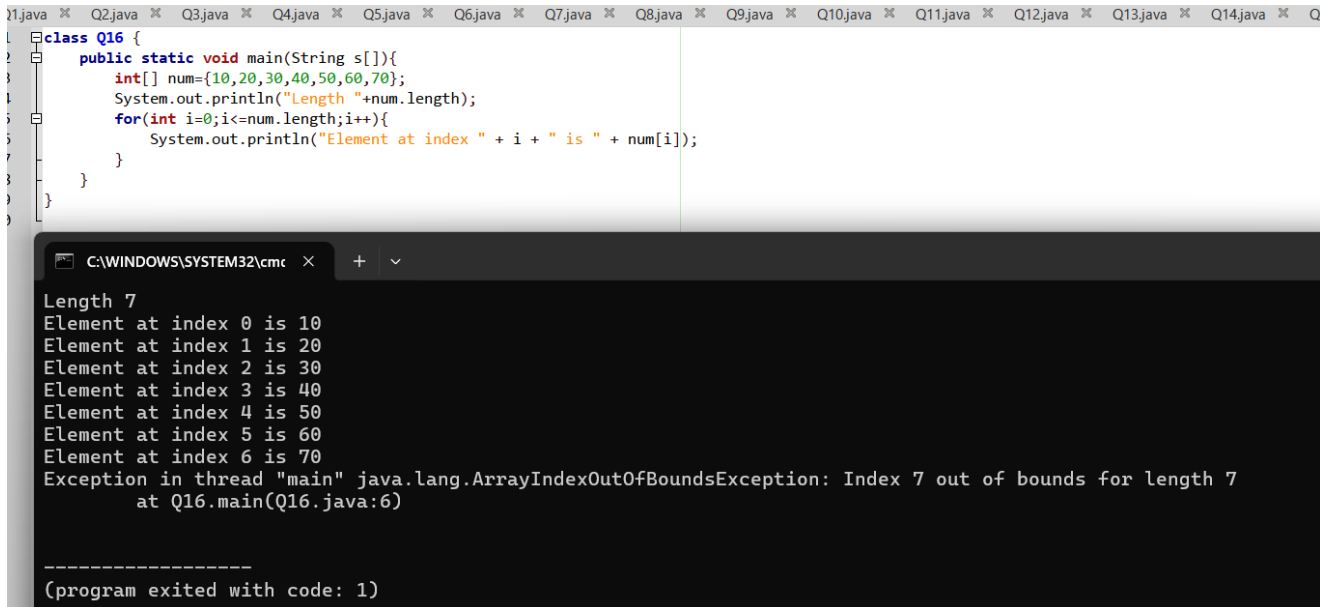
```
Q1.java ✕ Q2.java ✕ Q3.java ✕ Q4.java ✕ Q5.java ✕ Q6.java ✕ Q7.java ✕ Q8.java ✕ Q9.java ✕ Q10.java ✕
2 class Q15 {
3     public static void main(String[] args){
4         Scanner input=new Scanner(System.in);
5         Random num = new Random();
6         System.out.print("Enter The size of array : ");
7         int size=input.nextInt();
8
9         int[] ar= new int[size];
10        for(int i=0;i<ar.length;i++){
11            ar[i]=num.nextInt(100)+1;
12        }
13        System.out.println(Arrays.toString(ar));
14
15        int odd_count=0;
16        for(int i=0;i<ar.length;i++){
17            if(i%2==1){
18                if(ar[i]%2==1){
19                    odd_count++;
20                }
21            }
22        }
23        System.out.println("Count of how many odd numbers are in the odd index of array : "+odd_count);
24        boolean found100 =false;
25        for(int i=0;i<ar.length;i++){
26            if(ar[i]==100){
27                found100=true;
28                break;
29            }
30        }
31        System.out.println("The number 100 is present in the array : "+(found100 ? "Yes":"No"));
32
33        int max=0;
34        int min=ar[0];
35
36        for(int i=0;i<ar.length;i++){
37            if(ar[i]>max){
38                max=ar[i];
39            }
40        }
41        System.out.println("The maximum number of array : "+max);
42
43        for(int i=1;i<ar.length;i++){
44            if(ar[i]<=min){
45                min=ar[i];
46            }
47        }
48        System.out.println("The minimum number of array : "+min);
49    }
50 }
51
```

Output>>

```
Enter The size of array : 8
[69, 45, 94, 70, 84, 51, 76, 26]
Count of how many odd numbers are in the odd index of array : 2
The number 100 is present in the array : No
The maximum number of array : 94
The minimum number of array : 26
```

Q16.

i.



```
class Q16 {  
    public static void main(String s[]){  
        int[] num={10,20,30,40,50,60,70};  
        System.out.println("Length "+num.length);  
        for(int i=0;i<=num.length;i++){  
            System.out.println("Element at index " + i + " is " + num[i]);  
        }  
    }  
}
```

C:\WINDOWS\SYSTEM32\cmd

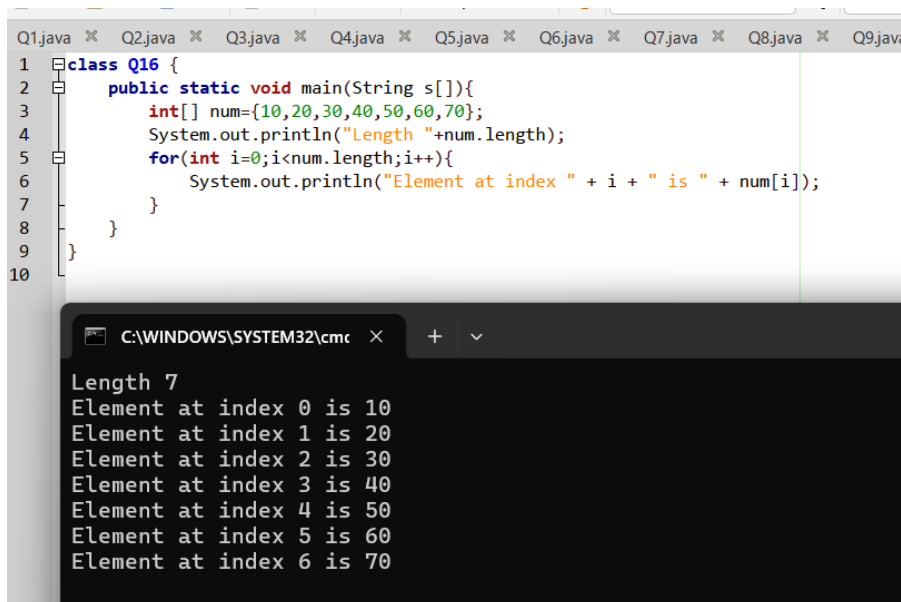
Length 7
Element at index 0 is 10
Element at index 1 is 20
Element at index 2 is 30
Element at index 3 is 40
Element at index 4 is 50
Element at index 5 is 60
Element at index 6 is 70
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 7 out of bounds for length 7
 at Q16.main(Q16.java:6)

(program exited with code: 1)

i Explain

We can't use `i<=num.length` Because that Array length is 7 so valid indices are only 0 to 6.when `i=7` ,accessing `num[7]` throws an `ArrayIndexOutOfBoundsException` .

ii.



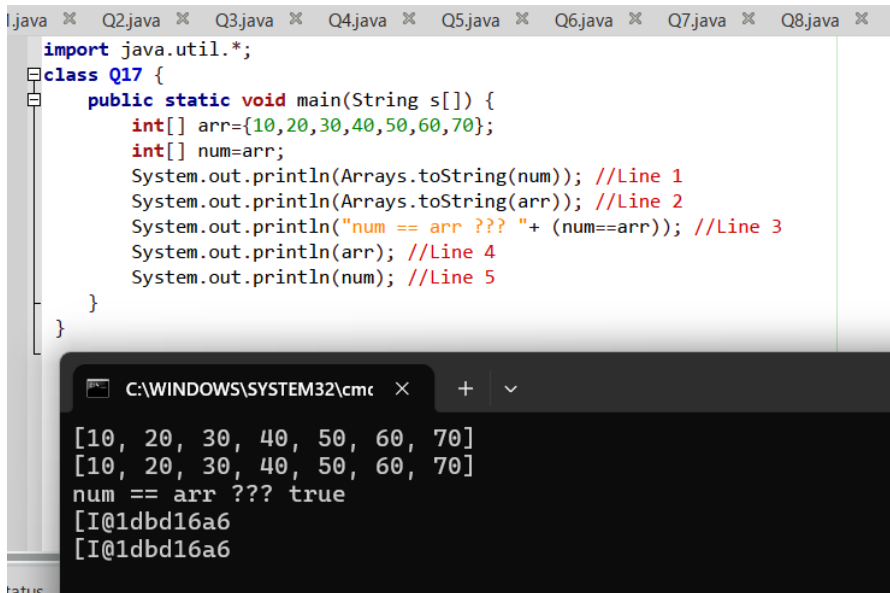
```
class Q16 {  
    public static void main(String s[]){  
        int[] num={10,20,30,40,50,60,70};  
        System.out.println("Length "+num.length);  
        for(int i=0;i<num.length;i++){  
            System.out.println("Element at index " + i + " is " + num[i]);  
        }  
    }  
}
```

C:\WINDOWS\SYSTEM32\cmd

Length 7
Element at index 0 is 10
Element at index 1 is 20
Element at index 2 is 30
Element at index 3 is 40
Element at index 4 is 50
Element at index 5 is 60
Element at index 6 is 70

Q17.

i,ii answers



```
import java.util.*;
class Q17 {
    public static void main(String s[]) {
        int[] arr={10,20,30,40,50,60,70};
        int[] num=arr;
        System.out.println(Arrays.toString(num)); //Line 1
        System.out.println(Arrays.toString(arr)); //Line 2
        System.out.println("num == arr ??? "+ (num==arr)); //Line 3
        System.out.println(arr); //Line 4
        System.out.println(num); //Line 5
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
[10, 20, 30, 40, 50, 60, 70]
[10, 20, 30, 40, 50, 60, 70]
num == arr ??? true
[I@1dbd16a6
[I@1dbd16a6
```

Explain i:

Line3:and after the `int num=arr;` these are not different these are same now then that is why line 3 boolean logic get output as true

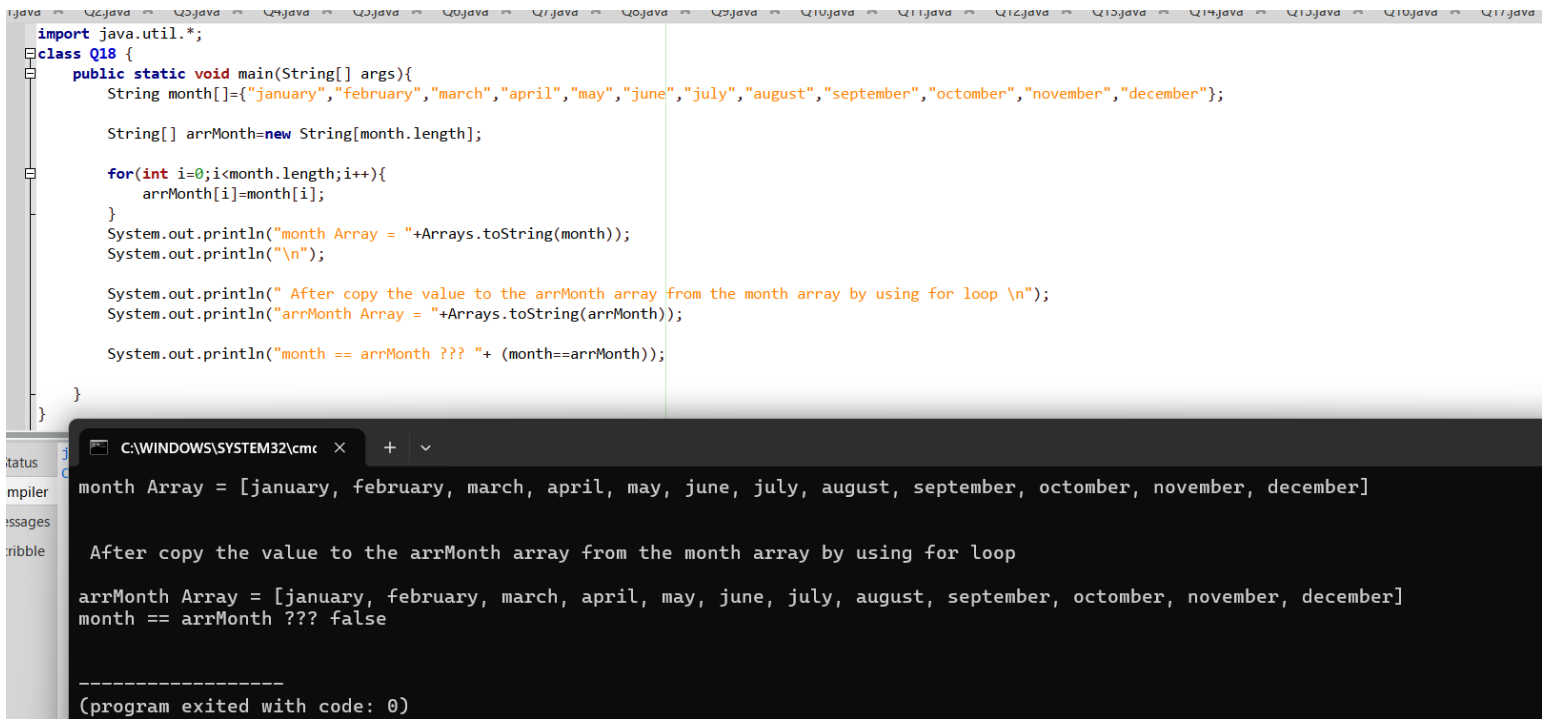
Line4: we already references arr array to num array variables arr array give his ram address as reference and that is why line 4 get output as `I@1dbd16a6`

Explain ii:

after assigned the reference of arr to num that line 5 `System.out.println` give output as `I@1dbd16a6`

Q18.

i,ii answers



```
import java.util.*;
class Q18 {
    public static void main(String[] args){
        String month[]={"january","february","march","april","may","june","july","august","september","octomber","november","december"};

        String[] arrMonth=new String[month.length];

        for(int i=0;i<month.length;i++){
            arrMonth[i]=month[i];
        }
        System.out.println("month Array = "+Arrays.toString(month));
        System.out.println("\n");

        System.out.println(" After copy the value to the arrMonth array from the month array by using for loop \n");
        System.out.println("arrMonth Array = "+Arrays.toString(arrMonth));

        System.out.println("month == arrMonth ??? "+ (month==arrMonth));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
month Array = [january, february, march, april, may, june, july, august, september, octomber, november, december]

 After copy the value to the arrMonth array from the month array by using for loop

arrMonth Array = [january, february, march, april, may, june, july, august, september, octomber, november, december]
month == arrMonth ??? false

-----
(program exited with code: 0)
```

ii explain: **(String[] arrMonth = new String[month.length];)**: Uses new keyword. Creates a **different** array object in Heap memory with a new address and I give output as **false**.

Q19.

i,ii output

```
i= C:\WINDOWS\SYSTEM32\cmd
[i] [3, 1, 3, 13, 2, 9, -3, -1, -4, 7, 12, -5, 10, -2, 7, -2, -1, -3, -1, 4, 0, 14, 9, -4, 11, 4, 1, 14, 14, 6, -3, 13, 5, 13, 6, -3, 2, 1, 8, 1, 14, 7, 10, 13, 1, 13, 4, -5, 8, -5]
out [-5, 8, -5, 4, 13, 1, 13, 10, 7, 14, 1, 8, 1, 2, -3, 6, 13, 5, 13, -3, 6, 14, 14, 1, 4, 11, -4, 9, 14, 0, 4, -1, -3, -1, -2, 7, -2, 10, -5, 12, 7, -4, -1, -3, 9, 2, 13, 3, 1, 3]
out
i=
t= The probability of getting negative numbers as Random numbers : 0.280
[i]
[j] (program exited with code: 0)
Press any key to continue . . .
```

Q20.

```
Q2.java x Q3.java x Q4.java x Q5.java x Q6.java x Q7.java x Q8.java x
import java.util.*;
class Q20 {
    public static void main(String[] args) {
        char vowels1[] = {'A', 'E', 'I', 'O', 'U'};
        char vowels2[] = {'a', 'e', 'i', 'o', 'u'};

        char[] temp1 = new char[vowels1.length + vowels2.length];

        for (int i = 0; i < vowels1.length; i++) {
            temp1[i] = vowels1[i];
        }
        for (int i = 0; i < vowels2.length; i++) {
            temp1[vowels1.length + i] = vowels2[i];
        }
        System.out.println(Arrays.toString(temp1));
        char[] temp2 = new char[vowels1.length + vowels2.length];

        for (int i = 0; i < vowels1.length; i++) {
            temp2[i * 2] = vowels1[i];
            temp2[i * 2 + 1] = vowels2[i];
        }
        System.out.println(Arrays.toString(temp2));

        char[] temp3 = new char[vowels1.length + vowels2.length];

        for (int i = 0; i < vowels1.length; i++) {
            temp3[i * 2] = vowels1[vowels1.length - 1 - i];
            temp3[i * 2 + 1] = vowels2[vowels2.length - 1 - i];
        }
        System.out.println(Arrays.toString(temp3));
        char[] temp4 = new char[vowels1.length + vowels2.length];

        for (int i = 0; i < vowels1.length; i++) {
            temp4[i * 2] = vowels1[i];
            temp4[i * 2 + 1] = vowels2[vowels2.length - 1 - i];
        }
        System.out.println(Arrays.toString(temp4));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
[A, E, I, O, U, a, e, i, o, u]
[A, a, E, e, I, i, O, o, U, u]
[U, u, O, o, I, i, E, e, A, a]
[A, u, E, o, I, i, O, e, U, a]
```

Q21.

```
21. class Demo{  
    public static void main(String s[]) {  
        int[] ar1 = new int[10];  
        double[] dr1 = new double[10];  
        //new line  
    }  
}
```

Which of the following assignment can be inserted at a new line, and still code will compile?

- | | | |
|---------------------|-----------------------|---------------------|
| A) dr1 = ar1; | B) ar1 = dr1; | C) dr1[0] = ar1[0]; |
| D) ar1[0] = dr1[0]; | E) ar1[0]=(int)dr1[0] | |

Answers

A,C,D,E

Q22.

Answer A

Q23.

23. Which can be inserted at line 20, still the code will compile?

```
class Example{  
    public static void main(String args[]){  
        int x=0;  
        int[] xr=new int[3];  
        double d=0.0;  
        double[] dr=new double[5];  
        int[] grade={'a','b'};  
        //Insert code here //Line 20  
    }  
}
```

- | | | |
|----------------------|-----------------|---------------------|
| A. x=xr[0]; | B. xr[0]=x; | C. x=xr; |
| D. xr=x; | E. dr[0]=xr[0]; | F. xr[0]=dr[0]; |
| G. xr[0]=(int)dr[0]; | H. xr=dr; | I. dr=(double[])xr; |
| J. dr=xr; | K. xr=(int)dr; | L. xr=(int[])dr; |

Answers A,B,E,G

Q24.

```
Q4.java x Q5.java x Q6.java x Q7.java x Q8.java x Q9.java x Q10.java x Q11.java x Q12.java x Q13.j
import java.util.*;
class Demo{
    public static void increment(int salary1, int salary2, int salary3, int salary4){
        salary1 +=5000;
        salary2 +=5000;
        salary3 +=5000;
        salary4 +=5000;
    }
    public static void main(String args[]){
        double[] salary={71,86,54,89};
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
        increment(salary [0], salary [1], salary [2], salary [3]); //Line 2
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
    }
}
```

javac "Q24.java" (in directory: D:\Rush\ICD124\day07)

Q24.java:12: error: incompatible types: possible lossy conversion from double to int
increment(salary [0], salary [1], salary [2], salary [3]); //Line 2
^

Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output
1 error
Compilation failed.

- i. We can't run this without type casting

ii.

```
Q4.java × Q5.java × Q6.java × Q7.java × Q8.java × Q9.java × Q10.java × Q11.java × Q12.java × Q13.java ×
import java.util.*;
class Q24 {
    public static void increment(double salary1, double salary2, double salary3, double salary4){
        salary1 +=5000;
        salary2 +=5000;
        salary3 +=5000;
        salary4 +=5000;
    }
    public static void main(String args[]){
        double[] salary={71,86,54,89};
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
        increment(salary [0], salary [1], salary [2], salary [3]); //Line 2
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
71.0 86.0 54.0 89.0
71.0 86.0 54.0 89.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

iii.

```
Q4.java × Q5.java × Q6.java × Q7.java × Q8.java × Q9.java × Q10.java × Q11,
1 import java.util.*;
2 class Q24 {
3     public static void increment(double[] salary){
4         for(int i=0;i<salary.length;i++){
5             salary[i] +=5000;
6         }
7     }
8 }
9 public static void main(String args[]){
10     double[] salary={71,86,54,89};
11     for(int i=0;i<salary.length;i++){
12         System.out.print(salary [i]+" ");
13     }
14     System.out.println();
15     increment(salary);
16     for(int i=0;i<salary.length;i++){
17         System.out.print(salary [i]+" ");
18     }
19 }
20 }
21 }
```

```
C:\WINDOWS\SYSTEM32\cmd
71.0 86.0 54.0 89.0
5071.0 5086.0 5054.0 5089.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q25.

```
Q5.java Q6.java Q7.java Q8.java Q9.java Q10.java Q11.java Q12.java
1  import java.util.*;
2  class Q25 {
3      public static void main(String[] args){
4          int dollar[]={15000, 21000, 18500, 12275, 35550, 28760, 17175, 10500};
5          double[] rupee = new double[dollar.length];
6
7          for(int i = 0; i < rupee.length; i++){
8              rupee[i] = (double)dollar[i] * 360.10;
9          }
10
11         int longvalue = 7500000;
12         int more_count = 0;
13         int less_count = 0;
14
15         for(int i = 0; i < rupee.length; i++){
16             if(rupee[i] > longvalue){
17                 more_count++;
18             } else {
19                 less_count++;
20             }
21         }
22
23         System.out.println("Count of Software engineers get more than Rs.7,500,000 : " + more_count);
24         System.out.println("Count of Software engineers get less than Rs.7,500,000 : " + less_count);
25
26         double[] arrmore = new double[more_count];
27         double[] arrless = new double[less_count];
28
29         int indexMore = 0;
30         int indexLess = 0;
31
32         for (int i = 0; i < rupee.length; i++) {
33             if (rupee[i] > 7500000) {
34                 arrmore[indexMore++] = rupee[i];
35             } else if (rupee[i] < 7500000) {
36                 arrless[indexLess++] = rupee[i];
37             }
38         }
39
40         System.out.print("Arrmore: ");
41         for (double val : arrmore) {
42             System.out.printf("%.2f ", val);
43         }
44         System.out.println();
45
46         System.out.print("Arrless: ");
47         for (double val : arrless) {
48             System.out.printf("%.2f ", val);
49         }
50         System.out.println();
51     }
52 }
53
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd
Q25.
i
c
Count of Software engineers get more than Rs.7,500,000 : 3
Count of Software engineers get less than Rs.7,500,000 : 5
Arrmore: 7562100.00 12801555.00 10356476.00
Arrless: 5401500.00 6661850.00 4420227.50 6184717.50 3781050.00

-----
(program exited with code: 0)
```

Q26.

```
Q6.java x Q7.java x Q8.java x Q9.java x Q10.java x Q11.java
import java.util.*;
class Q26 {
    public static void sort(int[] ar){
        for(int j=0; j<3; j++){ //loop1
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        for(int j=0; j<3; j++){ //loop2
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        for(int j=0; j<3; j++){ //loop3
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        System.out.print(Arrays.toString(ar));
    }
    public static void main(String[] args){
        int arr[]={96,17,54,6,81,47,63,79,32,13};
        sort(arr);
    }
}
```

i.

```
C:\WINDOWS\SYSTEM32\cmd.exe
[6, 17, 54, 96, 81, 47, 63, 79, 32, 13]
-----
(program exited with code: 0)
```

ii.

Loop 1: It compares adjacent pairs among the first 4 elements and pushes the largest value (96) to index 3.

- *Result:* [17, 54, 6, 96, 81, 47, 63, 79, 32, 13]

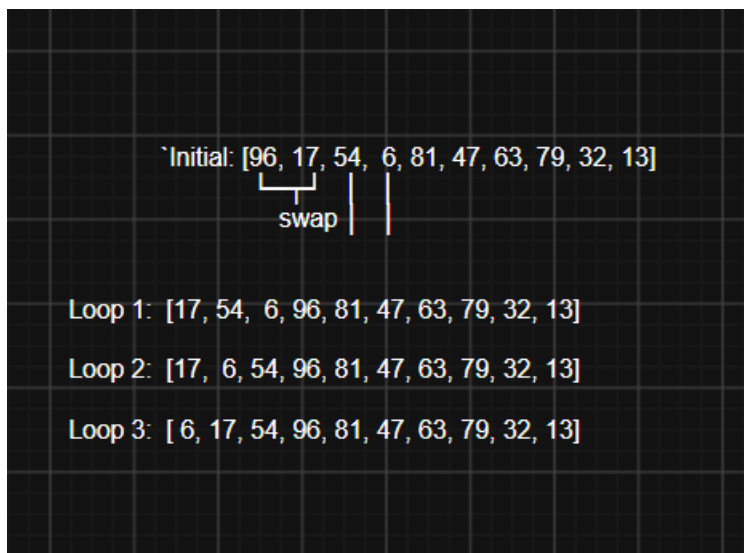
Loop 2: It runs again on the first 4 elements and pushes the second largest value (54) to index 2.

- *Result:* [17, 6, 54, 96, 81, 47, 63, 79, 32, 13]

Loop 3: It runs a final time, moving the smallest value (6) to the front at index 0.

- *Result:* [6, 17, 54, 96, 81, 47, 63, 79, 32, 13]

iii.



iv.

```
Q6.java x Q7.java x Q8.java x Q9.java x Q10.java x Q11.java x
import java.util.*;
class Q26 {
    public static void sort(int[] ar) {
        for (int i = 0; i < ar.length - 1; i++) {
            for (int j = 0; j < ar.length - 1 - i; j++) {
                if (ar[j] > ar[j + 1]) {
                    int t = ar[j];
                    ar[j] = ar[j + 1];
                    ar[j + 1] = t;
                }
            }
        }
        System.out.println(Arrays.toString(ar));
    }

    public static void main(String[] args) {
        int[] arr = {96, 17, 54, 6, 81, 47, 63, 79, 32, 13};
        sort(arr);
    }
}

C:\WINDOWS\SYSTEM32\cmd.exe x + v
[6, 13, 17, 32, 47, 54, 63, 79, 81, 96]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q27.

```
Q7.java × Q8.java × Q9.java × Q10.java × Q11.java × Q12.java × Q13.java × Q14.java × Q15.java × Q16.j
import java.util.*;
class Q27 {
    public static void sortSalaries(double[] salaries) {
        for (int i = 0; i < salaries.length - 1; i++) {
            for (int j = 0; j < salaries.length - 1 - i; j++) {
                if (salaries[j] > salaries[j + 1]) {
                    double temp = salaries[j];
                    salaries[j] = salaries[j + 1];
                    salaries[j + 1] = temp;
                }
            }
        }
    }

    public static void main(String[] args) {
        double[] dollarSalaries = {15000, 21000, 18500, 12275, 35550, 28760, 17175, 10500};

        sortSalaries(dollarSalaries);

        System.out.println(Arrays.toString(dollarSalaries));
    }
}

C:\WINDOWS\SYSTEM32\cmd × + ▾
[10500.0, 12275.0, 15000.0, 17175.0, 18500.0, 21000.0, 28760.0, 35550.0]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q28.

```
Q8.java × Q9.java × Q10.java × Q11.java × Q12.java × Q13.java × Q14.java × Q15.java × Q16.java ×
import java.util.*;
class Q28 {
    public static int[] modifyArray(int[] inputArr, int replaceIndex, int replaceValue) {
        int[] newArr = new int[inputArr.length];

        for (int i = 0; i < inputArr.length; i++) {
            newArr[i] = inputArr[i];
        }

        if (replaceIndex >= 0 && replaceIndex < newArr.length) {
            newArr[replaceIndex] = replaceValue;
        } else {
            System.out.println("Invalid index!");
        }

        return newArr;
    }

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int[] originalArray = {10, 20, 30, 40};

        System.out.print("Input array -> ");
        System.out.println(Arrays.toString(originalArray));

        System.out.print("Replace index - ");
        int index = input.nextInt();

        System.out.print("Replace value - ");
        int value = input.nextInt();

        int[] newArray = modifyArray(originalArray, index, value);

        System.out.print("New array -> ");
        System.out.println(Arrays.toString(newArray));
    }
}
```

C:\WINDOWS\SYSTEM32\cmd × + ▾

```
Input array -> [10, 20, 30, 40]
Replace index - 3
Replace value - 60
New array -> [10, 20, 30, 60]
```

Q29.

```
Q9.java × Q10.java × Q11.java × Q12.java × Q13.java × Q14.java × Q15.java × Q16.java × Q17.java × Q18.java × Q19.java × Q20.java × Q24.java × Q25.java × Q26.java × Q27.java ×
import java.util.*;
class Q29 {
    public static void main(String[] args){
        String subject[] = {"Mathematics", "Science", "English", "Sinhala", "History", "Buddhisum", "Business Studies", "Computer Studies"};

        String[] temp = new String[subject.length + 1];

        for (int i = 0; i < temp.length; i++) {
            if (i < 7) {
                temp[i] = subject[i];
            } else if (i == 7) {
                temp[i] = "Music";
            } else {
                temp[i] = subject[i - 1];
            }
        }

        System.out.println(Arrays.toString(temp));
    }
}
```

C:\WINDOWS\SYSTEM32\cmd × + ▾

[Mathematics, Science, English, Sinhala, History, Buddhisum, Business Studies, Music, Computer Studies]

(program exited with code: 0)

Press any key to continue . . . |

Q30.

```
Q10.java × Q11.java × Q12.java × Q13.java × Q14.java × Q15.java × Q16.java × Q17.java × Q18.java × Q19.java × Q20.java × Q24.java × Q25.java × Q26.j
import java.util.*;
class Q30 {
    public static void sortDescending(String[] months, int[] reserves) {
        for (int i = 0; i < reserves.length - 1; i++) {
            for (int j = 0; j < reserves.length - 1 - i; j++) {
                if (reserves[j] < reserves[j + 1]) {
                    int tempReserve = reserves[j];
                    reserves[j] = reserves[j + 1];
                    reserves[j + 1] = tempReserve;

                    String tempMonth = months[j];
                    months[j] = months[j + 1];
                    months[j + 1] = tempMonth;
                }
            }
        }
    }

    public static void main(String[] args) {
        String[] originalMonths = {"January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"};
        int[] originalReserves = {2362, 2311, 1917, 1812, 1887, 1854, 1817, 1717, 1779, 1705, 1806, 1896};

        String[] newMonths = new String[originalMonths.length];
        int[] newReserves = new int[originalReserves.length];

        for (int i = 0; i < originalMonths.length; i++) {
            newMonths[i] = originalMonths[i];
            newReserves[i] = originalReserves[i];
        }

        System.out.println("According to Month Order:");
        for (int i = 0; i < originalMonths.length; i++) {
            System.out.println(originalMonths[i] + " - " + originalReserves[i]);
        }

        System.out.println();

        sortDescending(newMonths, newReserves);

        System.out.println("According to Descending Order of Reserves:");
        for (int i = 0; i < newMonths.length; i++) {
            System.out.println(newMonths[i] + " - " + newReserves[i]);
        }
    }
}
```

```
impc
clas
```

According to Month Order:

January - 2362
February - 2311
March - 1917
April - 1812
May - 1887
June - 1854
July - 1817
August - 1717
September - 1779
October - 1705
November - 1806
December - 1896

According to Descending Order of Reserves:

January - 2362
February - 2311
March - 1917
December - 1896
May - 1887
June - 1854
July - 1817
April - 1812
November - 1806
September - 1779
August - 1717
October - 1705

(program exited with code: 0)

Q31.

```
Q11.java × Q12.java × Q13.java × Q14.java × Q15.java × Q16.java × Q17.java × Q18.java × Q19.java
1  import java.util.*;
2  class Q31 {
3      public static void main(String[] args) {
4          String[] names = {"Sachini", "Tharusha", "Shehan", "Ashini", "Sachin", "Kasun", "Lihini", "Hashini", "Ashini", "Kaveesha"};
5          int[] marks = {807, 796, 728, 801, 818, 746, 781, 756, 801, 812};
6
7          System.out.println("--- Part I ---");
8          for (int i = 0; i < names.length; i++) {
9              System.out.println(names[i] + " - " + marks[i]);
10         }
11         names = Arrays.copyOf(names, names.length + 1);
12         marks = Arrays.copyOf(marks, marks.length + 1);
13         names[names.length - 1] = "Amesh";
14         marks[marks.length - 1] = 804;
15
16         System.out.println("\n--- Part II ---");
17         for (int i = 0; i < names.length; i++) {
18             System.out.println(names[i] + " - " + marks[i]);
19         }
20         String[] newNames = new String[names.length - 1];
21         int[] newMarks = new int[marks.length - 1];
22         for (int i = 0, j = 0; i < names.length; i++) {
23             if (i != 8) {
24                 newNames[j] = names[i];
25                 newMarks[j] = marks[i];
26                 j++;
27             }
28         }
29         names = newNames;
30         marks = newMarks;
31
32         System.out.println("\n--- Part III ---");
33         for (int i = 0; i < names.length; i++) {
34             System.out.println(names[i] + " - " + marks[i]);
35         }
36         for (int i = 0; i < marks.length - 1; i++) {
37             for (int j = 0; j < marks.length - 1 - i; j++) {
38                 if (marks[j] < marks[j + 1]) {
39                     int tempM = marks[j];
40                     marks[j] = marks[j + 1];
41                     marks[j + 1] = tempM;
42
43                     String tempN = names[j];
44                     names[j] = names[j + 1];
45                     names[j + 1] = tempN;
46                 }
47             }
48         }
49         System.out.println("\n--- Part IV ---");
50         for (int i = 0; i < names.length; i++) {
51             System.out.println(names[i] + " - " + marks[i]);
52         }
53     }
54 }
```

```
i --- Part I ---
{ Sachini - 807
7 Tharusha - 796
n Shehan - 728
i Ashini - 801
Kasun - 818
p Kasun - 746
p Lihini - 781
h Hashini - 756
h Ashini - 801
Kaveesha - 812
n
--- Part II ---
i Sachini - 807
Tharusha - 796
n Shehan - 728
Ashini - 801
Sachin - 818
j Kasun - 746
j Lihini - 781
Hashini - 756
Ashini - 801
Kaveesha - 812
Amesh - 804
n
--- Part III ---
i Sachini - 807
Tharusha - 796
0 Shehan - 728
[ Ashini - 801
[ Sachin - 818
[ Kasun - 746
[ Lihini - 781
g Hashini - 756
[ Kaveesha - 812
[ Amesh - 804
n
--- Part IV ---
n Sachin - 818
Kaveesha - 812
i Sachini - 807
Amesh - 804
Ashini - 801
Tharusha - 796
Lihini - 781
Hashini - 756
Kasun - 746
Shehan - 728
```

Q32.

```
< Q12.java ✕ Q13.java ✕ Q14.java ✕ Q15.java ✕ Q16.java ✕ Q17.java
1  import java.util.*;
2  class Q32 {
3      public static void rotateLeft(int[] arr) {
4          int first = arr[0];
5          for (int i = 0; i < arr.length - 1; i++) {
6              arr[i] = arr[i + 1];
7          }
8          arr[arr.length - 1] = first;
9      }
10
11     public static void main(String[] args) {
12         int[] arr = {10, 20, 30, 40, 50, 60, 70};
13         System.out.println(Arrays.toString(arr));
14         rotateLeft(arr);
15
16         System.out.println(Arrays.toString(arr));
17     }
18 }
19
```

```
C:\WINDOWS\SYSTEM32\cmd ✕ + v
[10, 20, 30, 40, 50, 60, 70]
[20, 30, 40, 50, 60, 70, 10]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q33.

```
Q13.java Q14.java Q15.java Q16.java Q17.java Q18.java
1  import java.util.*;
2  class Q33 {
3      public static void main(String[] args) {
4          int[] arr = {1, 0, 1, 1, 0, 0, 1, 0, 1, 0};
5
6          int zeroIndex = 0;
7          for (int i = 0; i < arr.length; i++) {
8              if (arr[i] == 0) {
9                  int temp = arr[i];
10                 arr[i] = arr[zeroIndex];
11                 arr[zeroIndex] = temp;
12                 zeroIndex++;
13             }
14         }
15
16         System.out.println(Arrays.toString(arr));
17     }
18 }
19
```

```
C:\WINDOWS\SYSTEM32\cmd  +  v
[0, 0, 0, 0, 0, 1, 1, 1, 1, 1]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q34.

```
Q14.java × Q15.java × Q16.java × Q17.java × Q18.java × Q19.java × Q20.java × Q24.java × Q25.java × Q26.java × Q27.java
import java.util.*;
class Q34 {
    public static void main(String[] args) {
        String[] countries = {"Switzerland(2)", "Ivory Coast(1)", "Jamaica(2)", "United States(1)",
                              "United States(2)", "Great Britain", "Ivory Coast(2)", "Jamaica(3)", "Switzerland(1)", "Jamaica(1)"};
        double[] times = {11.11, 10.86, 10.76, 10.98, 11.11, 11.05, 11.14, 10.61, 11.06, 10.74};

        int gold = 0;
        for (int i = 1; i < times.length; i++) {
            if (times[i] < times[gold]) gold = i;
        }
        System.out.println("Part I: " + countries[gold] + " - " + times[gold] + "s");

        int newGold = (gold == 0) ? 1 : 0;
        for (int i = 0; i < times.length; i++) {
            if (i != gold && times[i] < times[newGold]) newGold = i;
        }
        System.out.println("Part II: " + countries[newGold] + " - " + times[newGold] + "s");

        String[] c = new String[countries.length - 1];
        double[] t = new double[times.length - 1];

        for (int i = 0, j = 0; i < countries.length; i++) {
            if (i != gold) {
                c[j] = countries[i];
                t[j] = times[i];
                j++;
            }
        }

        for (int i = 0; i < t.length - 1; i++) {
            for (int j = 0; j < t.length - 1 - i; j++) {
                if (t[j] > t[j + 1]) {
                    double tempT = t[j]; t[j] = t[j + 1]; t[j + 1] = tempT;
                    String tempC = c[j]; c[j] = c[j + 1]; c[j + 1] = tempC;
                }
            }
        }

        System.out.println("\nPart III:");
        for (int i = 0; i < c.length; i++) {
            System.out.println(c[i] + " - " + t[i] + "s");
        }
    }
}
```

Part I: Jamaica(3) - 10.61s
Part II: Jamaica(1) - 10.74s

Part III:

Jamaica(1) - 10.74s
Jamaica(2) - 10.76s
Ivory Coast(1) - 10.86s
United States(1) - 10.98s
Great Britain - 11.05s
Switzerland(1) - 11.06s
Switzerland(2) - 11.11s
United States(2) - 11.11s
Ivory Coast(2) - 11.14s

(program exited with code: 0)

Press any key to continue . . .

Q35.

```
Q15.java  Q16.java  Q17.java  Q18.java  Q19.java  Q20.java  Q24.java
import java.util.*;
class Q35 {
    public static void findDuplicate(int[] arr) {
        boolean found = false;
        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    System.out.println("duplicate \"" + arr[i] + "\"");
                    found = true;
                    break;
                }
            }
            if (found) break;
        }
        if (!found) {
            System.out.println("no duplicate");
        }
    }

    public static boolean equals(int[] a, int[] b) {
        if (a.length != b.length) {
            return false;
        }
        for (int i = 0; i < a.length; i++) {
            if (a[i] != b[i]) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        int[] ar = {10, 20, 30, 40, 50};
        int[] br = {10, 20, 10, 40, 50, 60};
        int[] cr = {30, 30, 30, 30};
        int[] dr = {40, 10, 50, 20, 30};
        int[] er = {10, 20, 30, 40, 50};
        int[] fr = {10, 20, 30, 40};

        findDuplicate(ar);
        findDuplicate(br);
        findDuplicate(cr);
        findDuplicate(dr);
        findDuplicate(er);
        findDuplicate(fr);

        System.out.println("ar==br: " + equals(ar, br));
        System.out.println("ar==cr: " + equals(ar, cr));
        System.out.println("ar==dr: " + equals(ar, dr));
        System.out.println("ar==er: " + equals(ar, er));
        System.out.println("ar==fr: " + equals(ar, fr));
    }
}
```

C:\WINDOWS\SYSTEM32\cmd

```
no duplicate
duplicate "10"
duplicate "30"
no duplicate
no duplicate
no duplicate
ar==br: false
ar==cr: false
ar==dr: false
ar==er: true
ar==fr: false
```

(program exited with

Press any key to cont

Q36.

```
Q16.java × Q17.java × Q18.java × Q19.java × Q20.java × Q24.java × Q25.java × Q26.java × Q27.java
import java.util.*;
class Q36 {
    public static void main(String[] args) {
        String[] c = {"Shanaka", "Dinesh", "Amesh", "Lahiru", "Dilshan", "Hashan", "Wenura",
                     "Randika", "Dasun", "Thisara", "Kasun", "Nisal", "Praveen"};

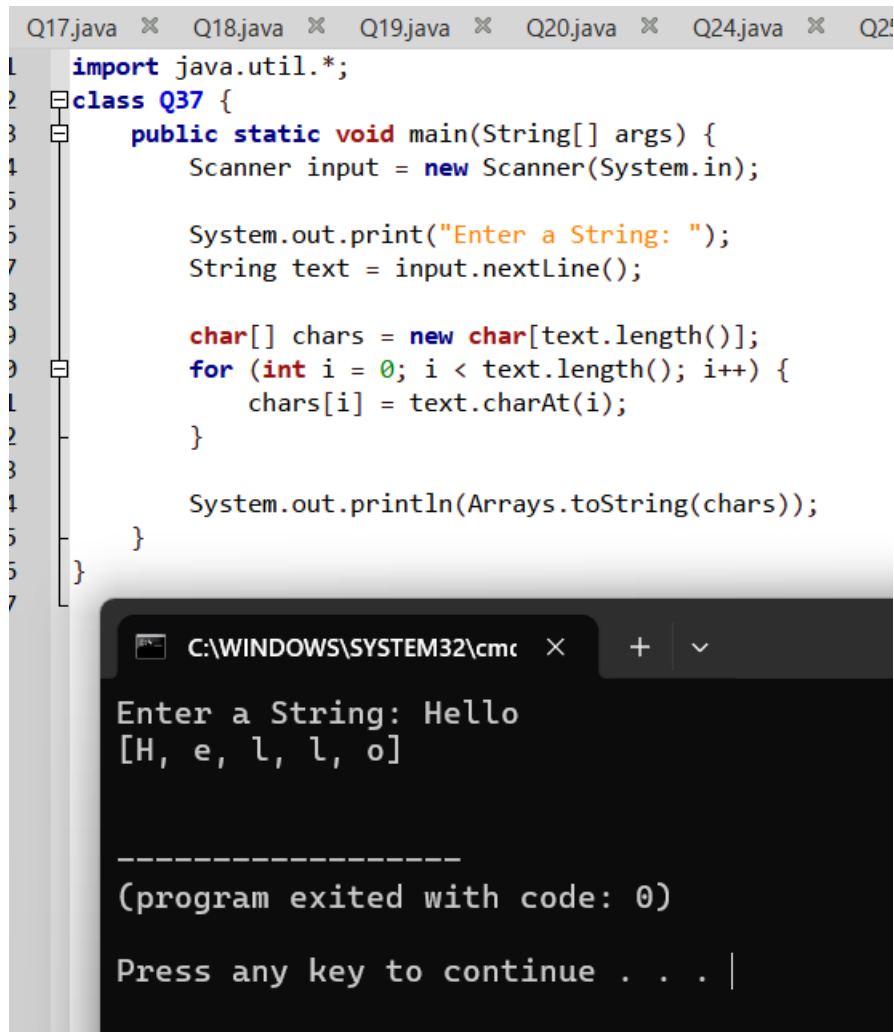
        String[] f = {"Anjana", "Chamara", "Kamal", "Dhanuka", "Shanaka", "Kasun", "Dineth",
                     "Sahan", "Wenura", "Amesh", "Mahesh", "Randika", "Thisara"};

        System.out.println("Part I:");
        for (String s1 : c) {
            for (String s2 : f) {
                if (s1.equals(s2)) {
                    System.out.println(s1);
                    break;
                }
            }
        }
        System.out.println("\nPart II:");
        String all = "";
        for (String s : c) {
            if (!all.contains(s)) {
                all += s + " ";
                System.out.println(s);
            }
        }
        for (String s : f) {
            if (!all.contains(s)) {
                all += s + " ";
                System.out.println(s);
            }
        }
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
Part I:
Shanaka
Amesh
Wenura
Randika
Thisara
Kasun

Part II:
Shanaka
Dinesh
Amesh
Lahiru
Dilshan
Hashan
Wenura
Randika
Dasun
Thisara
Kasun
Nisal
Praveen
Anjana
Chamara
Kamal
Dhanuka
Dineth
Sahan
Mahesh
```

Q37.



The image shows a Java IDE with several tabs open: Q17.java, Q18.java, Q19.java, Q20.java, Q24.java, and Q27.java. The active tab is Q37.java, which contains the following code:

```
1 import java.util.*;
2 class Q37 {
3     public static void main(String[] args) {
4         Scanner input = new Scanner(System.in);
5
6         System.out.print("Enter a String: ");
7         String text = input.nextLine();
8
9         char[] chars = new char[text.length()];
10        for (int i = 0; i < text.length(); i++) {
11            chars[i] = text.charAt(i);
12        }
13
14        System.out.println(Arrays.toString(chars));
15    }
16 }
```

Below the code editor, a command prompt window is open, showing the execution of the program. The prompt displays the input "Hello" and the resulting character array [H, e, l, l, o]. It also shows the message "(program exited with code: 0)" and a prompt to "Press any key to continue . . . |".

Q38.

```
Q18.java × Q19.java × Q20.java × Q24.java × Q25.java × Q26.java × Q27.java ×
import java.util.*;
class Q38 {
    public static void main(String[] args) {
        int x = 1;
        while (true) {
            if (hasSameDigits(x, 2 * x) &&
                hasSameDigits(x, 3 * x) &&
                hasSameDigits(x, 4 * x) &&
                hasSameDigits(x, 5 * x) &&
                hasSameDigits(x, 6 * x)) {

                System.out.println("Smallest integer x: " + x);
                break;
            }
            x++;
        }
    }
    public static boolean hasSameDigits(int n1, int n2) {
        char[] a1 = String.valueOf(n1).toCharArray();
        char[] a2 = String.valueOf(n2).toCharArray();
        Arrays.sort(a1);
        Arrays.sort(a2);
        return Arrays.equals(a1, a2);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd × + v
Smallest integer x: 142857

-----
(program exited with code: 0)

Press any key to continue . . .
```

Q39.

```
Q19.java Q20.java Q24.java Q25.java Q26.java Q27.java Q28.java Q29.java
import java.util.*;
class Q39 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int[] ids = new int[100];
        String[] names = new String[100];
        int count = 0;

        while (true) {
            System.out.print("Enter Admission Number: ");
            int id = input.nextInt();
            input.nextLine();

            boolean exists = false;
            for (int i = 0; i < count; i++) {
                if (ids[i] == id) {
                    exists = true;
                    break;
                }
            }

            if (exists) {
                System.out.println("Duplicate Admission number...");
                System.out.print("Do you want to enter another admission number? (y/n): ");
                if (input.nextLine().equalsIgnoreCase("n")) break;
                continue;
            }

            System.out.print("Enter Student Name: ");
            names[count] = input.nextLine();
            ids[count] = id;
            count++;

            System.out.print("Do you want to add another student? (y/n): ");
            if (input.nextLine().equalsIgnoreCase("n")) break;
        }

        System.out.println("\nRegistered Students:");
        for (int i = 0; i < count; i++) {
            System.out.println(ids[i] + " - " + names[i]);
        }
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
me
0
) Enter Admission Number: 567876
ou Enter Student Name: Kamal
= Do you want to add another student? (y/n): n
ex
e Registered Students:
t 567876 - Kamal
(i
e
b -----
(program exited with code: 0)

Press any key to continue . . . |
```

```
g[
ou Enter Admission Number: 4567
( Enter Student Name: Kasin
ys Do you want to add another student? (y/n): y
nt Enter Admission Number: 45678
np Enter Student Name: Ranutha
oo Do you want to add another student? (y/n): n
or

Registered Students:
4567 - Kasin
45678 - Ranutha

f -----
(program exited with code: 0)

Press any key to continue . . . |
```

Q40.

Q20.java Q24.java Q25.java Q26.java Q27.java Q28.java Q29.java Q30.java Q31.java

```
import java.util.*;

class Q40 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] ids = new int[1000];
        int count = 0;

        while (true) {
            System.out.print("Enter Clothing ID (-1 to exit): ");
            int id = input.nextInt();

            if (id == -1) break;

            boolean exists = false;
            for (int i = 0; i < count; i++) {
                if (ids[i] == id) {
                    exists = true;
                    break;
                }
            }

            if (exists) {
                System.out.println("Duplicate element! ID already exists.");
            } else {
                ids[count] = id;
                count++;
            }
        }

        int[] selected = new int[count];
        for (int i = 0; i < count; i++) {
            selected[i] = ids[i];
        }

        System.out.println("Selected Clothing IDs: " + Arrays.toString(selected));
    }
}
```

